THERMALOOP AI

Four Research Facilities Powered by Recovered Data-Center Heat

Building | Robotics & Materials Laboratory | Seed Center | Greenhouse



Figure 1. Conceptual campus rendering of a water-based waste-heat reuse network. Illustration only; not an as-built or dimensioned piping diagram.

Research scope

This technical note explains what recovered heat actually does in each of the four downstream facilities, which devices receive it, how hot-water supply and return are controlled, and which measurements are required to verify the benefit. It is a focused companion to the ThermaLoop AI digital-twin research manuscript.

Central engineering principle

GPU electrical power ultimately becomes heat. A liquid-cooling loop collects some of this heat; a heat exchanger transfers it to a separate secondary loop. A heat pump may raise the water temperature to a usable level. Control valves then allocate heat to independent building loads. Heat delivered to a facility is not automatically equal to heat recovered from the GPU cluster.

Color convention

RED / ORANGE = hot outgoing water and delivered thermal energy. BLUE / CYAN = cooler return water directed back to the plant. These colors describe temperature relative to the loop; they do not imply direct contact between server coolant and laboratory, potable-water, or greenhouse systems.

Status: proposed system architecture and illustrative calculations. No installation, operating cost, plant performance, or biological outcome has yet been experimentally validated.

1. Building Heating - Where the Heat Goes

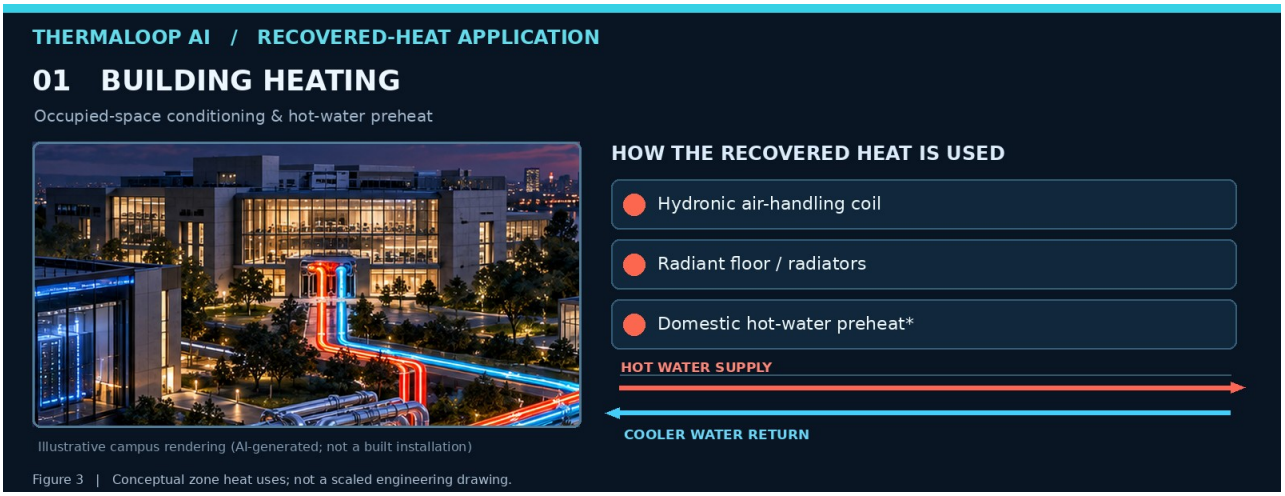


Figure 2. Building heat uses: air-handling-unit heating coil, hydronic radiant floor and separate domestic-hot-water preheating interface.

Function and equipment

The building uses the recovered heat primarily to maintain thermal comfort and reduce the output required from conventional heating equipment. A hot-water distribution circuit can supply air-handling-unit coils, radiators or radiant floors. Each load is isolated and regulated using mixing valves, flow-control valves, thermostats and return-temperature sensors.

- Space heating: a water-to-air coil warms air delivered through the ventilation system when the supply air would otherwise be too cold.
- Radiant floor or radiators: a hydronic circuit transfers heat into occupied rooms. Lower-temperature distribution is often advantageous when compatible with the building design.
- Potable-water preheating: an indirect, code-compliant heat exchanger can reduce the energy needed for domestic hot water. Final temperature and hygienic controls remain a separate responsibility.

How the control loop operates

A thermostat sets the room target temperature. A controller compares actual and required room temperature, opens the heating valve and adjusts the flow rate. When the room reaches its setpoint, the valve reduces flow. If recovered heat is insufficient, a backup heater provides the unmet load; if no space heating is required, the heating branch can be bypassed.

Variables to model and measure

Parameter	Meaning	Unit
Room / outdoor temperature	Boundary conditions for heating demand	°C
Supply / return water temperature	Thermal energy extracted by coil or floor	°C
Water mass flow	Hydronic branch throughput	kg/s
Delivered heat / unmet demand	Useful heat and backup requirement	kW

Example hypothesis: recovered heat may reduce auxiliary heating energy during cold-weather hours, but savings must be calculated from matched heating loads and actual heat-pump electricity.

2. Robotics & Materials Lab - Controlled Experimental Heating

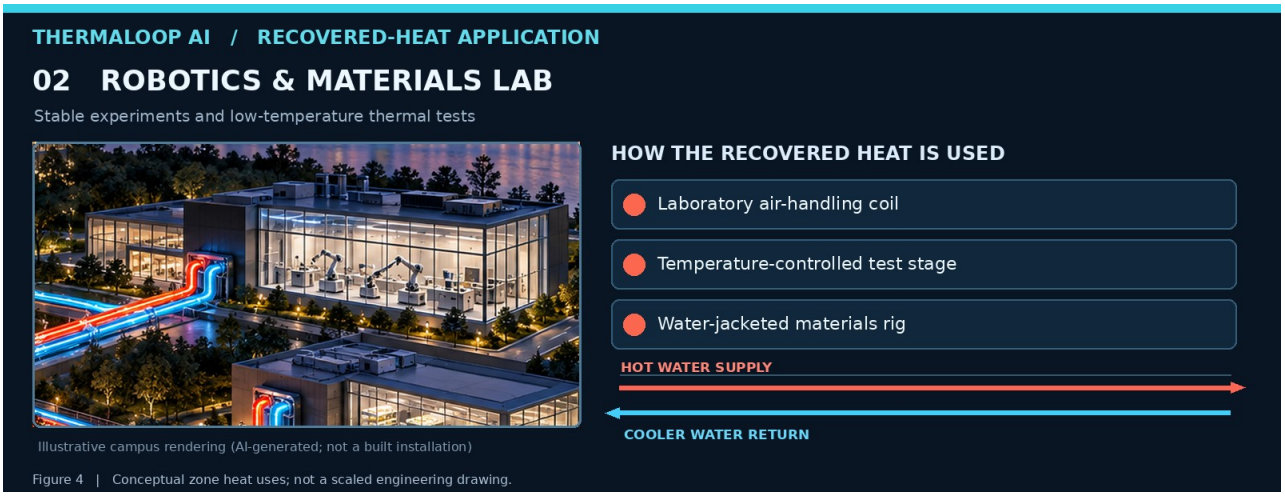


Figure 3. Laboratory heat uses: controlled room temperature and a dedicated, temperature-regulated experimental heating circuit.

Why a research laboratory needs heat

The primary requirement is a repeatable thermal environment. Temperatures affect polymer stiffness, viscoelasticity, calibration drift, actuator response and measurement uncertainty. Heat recovered from the data center can support room conditioning or a specialized low- to medium-temperature experimental apparatus, provided a separate safety-controlled interface is used.

- Ambient room control: a heating coil offsets outdoor heat losses and supports stable environmental conditions for robotics tests and materials measurements.
- Thermostated test rig: a secondary water-jacketed device or heat-transfer plate can expose materials, compliant mechanisms or soft actuators to defined temperatures.
- Preheating of suitable utilities: where the experimental protocol permits it, recovered heat can offset the thermal load of a separate process-heating system.

How tests should be designed

To isolate the effect of temperature, compare runs with the same sample, mechanical loading protocol, humidity, electronics and timing. Record the specimen temperature, ambient temperature, input electrical power, sensor calibration and output response. The water-loop controller must not introduce large thermal fluctuations that compromise the experiment.

Important boundary

The lab is not a direct destination for raw server coolant. Heat should cross a heat exchanger to an isolated laboratory circuit. The digital twin should explicitly track thermal setpoints and delivered thermal power, rather than assume that any arbitrary heat output is automatically useful.

Research output	Example measurement	Purpose
Experimental repeatability	Temperature stability over test period	Control confounding variables
Material response	Displacement, force, modulus vs temperature	Quantify temperature dependence
Heat service	kW delivered and kWh consumed	Check energy benefit

3. Seed Center - Germination and Storage Are Different

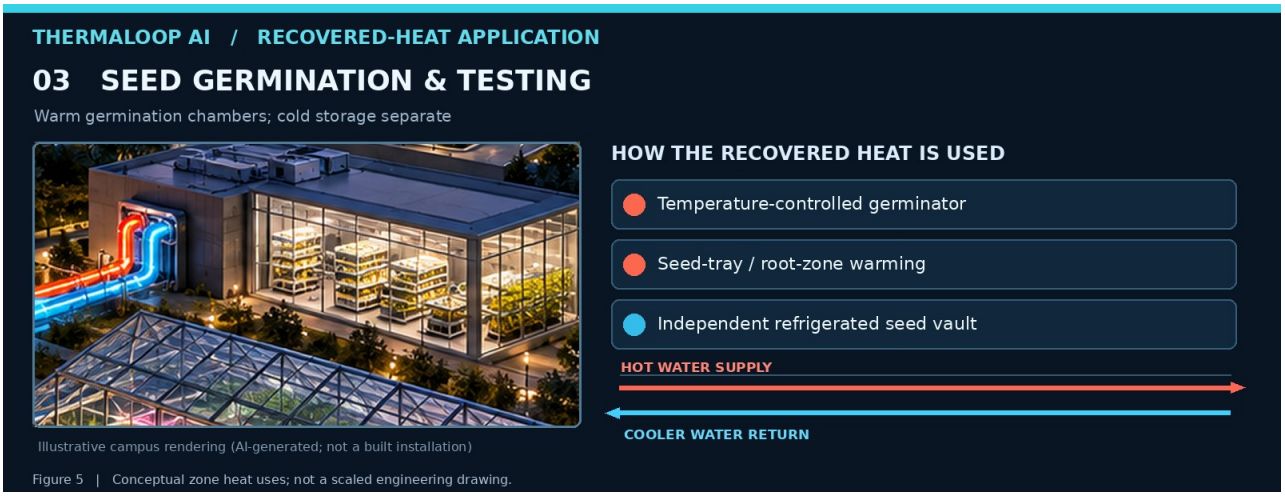


Figure 4. Seed center with a warm germination/test zone and a separately cooled, humidity-controlled long-term storage zone.

Warm functions: germination and experimental growth

Warm water is useful for seed germination when controlled substrate or chamber temperature improves conditions for a particular species. A heating coil, under-tray mat connected to a hydronic circuit, or insulated growth chamber can maintain a chosen setpoint. This makes heat recovery a possible input to controlled germination studies, not a substitute for biological protocols.

- Germination chambers: regulate air or substrate temperature for each experimental group without exposing seeds to uncontrolled hot pipes.
- Seed-tray / root-zone warming: maintain substrate temperature when room air is cooler, using temperature probes close to the seeds or roots.
- Comparison experiment: run groups under distinct species-appropriate setpoints and measure germination fraction, median germination time and early seedling vigor.

Cold function: seed storage must remain separate

Long-term seed preservation commonly requires cool or frozen and dry, stable conditions, depending on species and storage method. Such a vault must not be heated by the waste-heat loop. It requires independent refrigeration and humidity control, with clear thermal separation from the warm germination area. Refrigeration energy belongs in the total site energy budget.

Biological measurements and cautions

Variable	Use	Caution
Species and seed-lot identity	Controls biological variability	Do not assume one optimum temperature
Substrate/chamber temperature	Defines treatment	Measure near actual seed environment
Humidity / moisture	Supports comparable trials	Avoid mold or desiccation
Germination fraction and time	Measures outcome	Use adequate replication

Example: 20°C, 25°C and 30°C may be experimental treatments for suitable seeds, not universal recommended values. Choose setpoints using species-specific evidence.

4. Greenhouse - Delivering Heat Where Plants Need It

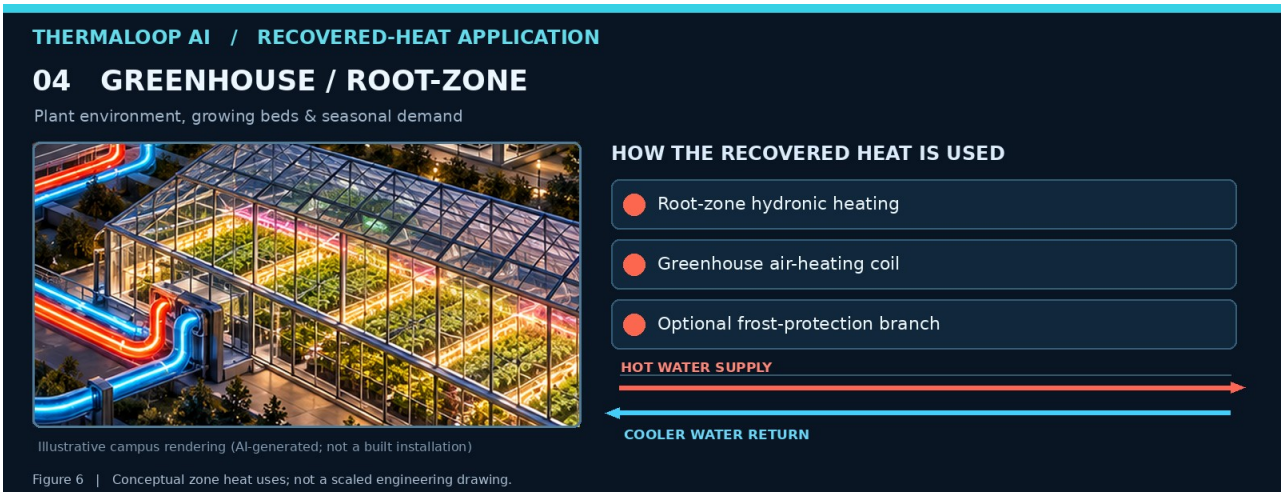


Figure 5. Greenhouse uses: hydronic air heating, root-zone warming and an independent cooler return line.

Functions of recovered hot water

A greenhouse can use recovered heat to stabilize the plant microclimate during cold periods. Heat may be delivered through hydronic pipe rails, coils, perimeter radiators or pipes below beds. The goal is to provide only the heating needed for plants and equipment, with attention to crop species, humidity, ventilation and condensation control.

- Root-zone heating: water pipes beneath beds or trays warm root substrate, sometimes allowing lower bulk-air temperature than a system that heats the entire volume.
- Air heating: a coil or distributed pipes compensate for greenhouse envelope and ventilation losses while maintaining the crop air-temperature setpoint.
- Irrigation-water preheating: an indirect exchanger can moderate water temperature when a crop-specific need is established; water should not be heated indiscriminately.
- Outdoor frost protection: selected nearby surfaces or pipes can be protected, but uncontrolled open-air heating is usually a high-loss load and needs separate cost justification.

Control and research questions

Use air temperature, substrate temperature, relative humidity, outdoor temperature and solar irradiation to decide when and where to deliver heat. Compare equal crop growth conditions while measuring the energy supplied to the root zone versus air heating. In summer, bypass or store heat when heating demand is absent; the server cooling system must remain safe.

Research objective	Sensor / metric	Decision variable
Maintain crop environment	Air and substrate °C, RH	Heating valve / loop flow
Minimize useful heat per crop output	Thermal kWh and plant yield proxy	Root vs air distribution
Avoid wasting heat	Greenhouse demand vs available kW	Bypass / storage / rejection

5. Engineering Calculations and Example Allocation

5.1 Heat transferred by a water branch

Useful heat delivered to a facility is calculated from the water mass-flow rate and the measured supply/return temperature difference:

$$Q_{zone}\ (kW) = \dot{m}\ (kg/s) \times c_p\ (kJ/kg\cdot K) \times [T_{supply} - T_{return}]\ (K)$$

For liquid water near typical heating temperatures, $c_p \approx 4.18$ kJ/(kg·K). If one branch receives 0.24 kg/s with a 10 K temperature drop, its delivered heat is approximately $0.24 \times 4.18 \times 10 = 10.0$ kW. The numbers are illustrative only.

5.2 Balance of recovered heat and building demand

A thermal control model should distinguish: (a) heat recovered from IT cooling, (b) useful heat after the heat pump, (c) piping and tank losses, (d) heat actually accepted by each zone, and (e) unserved heat demand.

$$Q_{available} + W_{heat\text{-}pump} \approx \Sigma\, Q_{delivered,zones} + Q_{losses} + Q_{rejected} + \Delta E_{storage}/\Delta t$$

The expression is an approximate heat balance across the chosen system boundary: $Q_{available}$ represents the heat-pump source-side heat, and $W_{heat\text{-}pump}$ is its electrical input. Take care not to add recovered source heat to heat-pump output twice.

5.3 Illustrative distribution scenario

Recipient	Heat demand (kW)	Possible service
Building	22	Space heating / HVAC
Robotics & Materials Lab	10	Room heating and controlled experiments
Seed Germination Center	6	Germination chambers only
Greenhouse	18	Air / root-zone heating
Network heat loss	4	Pipe and equipment losses
TOTAL	60	Illustrative balance

The 60 kW example is a notional instantaneous allocation. It is not a measurement from ThermaLoop AI, nor proof that a 64-GPU facility can supply these loads at all hours. For a 10 K design supply/return difference, supplying 56 kW across occupied loads would require approximately 1.34 kg/s total secondary-loop water flow.

5.4 Better network topology

For research-grade control, independent parallel branches from a common thermal header are generally preferable to a single pipe that visits all four facilities in strict sequence. A serial cascade is possible only where the temperature levels match, but downstream performance becomes coupled to upstream operation. The digital twin should expose this topology choice and track branch-specific valve position, temperatures, mass flow and unmet demand.

6. Measurement Plan, Controls and Research Boundaries

6.1 What each facility must report

Facility	Minimum sensors	Performance metric
Building	Supply/return water °C; flow; indoor/outdoor °C	Thermal kWh delivered; unmet comfort demand
Laboratory	Water °C/flow; room and specimen °C	Stability; valid experimental runs
Seed Center	Water °C/flow; chamber/substrate °C; humidity	Germination %, time; cold-vault energy
Greenhouse	Water °C/flow; air/substrate °C; RH; weather	Thermal kWh vs plant conditions

6.2 Supervisory control strategy

At each simulation timestep, the heat-plant controller estimates available heat and current demand in all four branches. It first enforces GPU and cooling safety limits; then serves critical temperature-controlled loads subject to thermal delivery constraints, heat-pump capacity, tank charge and backup heat availability. Any unserved demand is recorded explicitly. When no zone needs heat, the plant bypasses, stores or rejects surplus heat rather than forcing thermal delivery.

6.3 How to validate the claims

- Measure primary-loop heat removal and secondary-loop heat delivery independently; check the energy balance and sensor uncertainties.
- Use the same weather, occupancy, lab protocols, seed lots and greenhouse conditions when comparing baseline versus heat-reuse cases.
- Report electricity for the heat pump, secondary pumps, backup heater and refrigeration, not only the GPU cluster.
- Calculate heat actually used and financially displaced, not all theoretically recoverable heat.
- Publish operating hours, load profiles, calibrated assumptions and uncertainty intervals. Distinguish measured, calibrated and synthetic results.

Conclusion

The four facilities give ThermaLoop AI a credible research purpose only when heat-delivery mechanisms are made explicit. Building heating offsets comfort loads; laboratories require stable and isolated experimental heating; seed centers use heat for germination but not cold storage; and greenhouses support air and root-zone thermal management. A validated digital twin should connect these end uses to the AI cluster through a measured heat balance and independently controlled branch circuits.

Figure provenance: conceptual visualizations prepared for ThermaLoop AI. They are illustrative and not photographs or engineering drawings of installed infrastructure.